The motor gets connected across the supply mains through the main contactor. The motor continues to get supply even when the pressure on the push button is released. The contactor coil will then get supply through the sealing contact 'A' of the contactor.

Contactor 'A' is therefore called the hold on contact. When the 'STOP' push button is pressed, the coil gets de-energized, the main contacts of the contactor open and the motor stops. In case of over load on the motor, the contact of the over load relay (OLR) will open and subsequently the motor will stop. Fuses are provided for short circuit protection.

Advantages of DOL starter

i. Simple in construction.
ii. Easy to install.
iii. Easy to maintain.
iv. Inexpensive.

Disadvantages of DOL starter

i. Used only for fractional horse power motors.
ii. Starting current is high.
iii. Suitable only for the motors which attains rated speed in a very short time.

ii. Star-Delta method of starting

In this method, the stator-phase windings connected in 'Star' and full voltage is connected across its three terminals. As the motor picks up speed, the windings are disconnected through a switch and they are reconnected in 'Delta' across the supply terminals. The current drawn by the motor from the line is reduced to 1/3 as compared to the current it would have drawn if connected in delta.

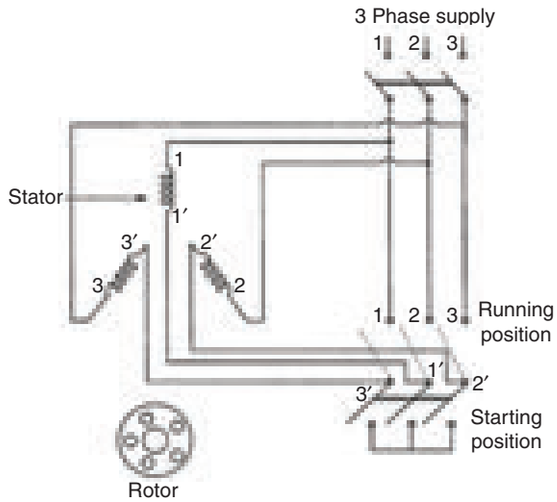


Fig. 8.4 Star delta starter

Reduce torque due to star connection

Torque developed by an induction motor is proportional to the square of the applied voltage. As the phase voltage is reduced to 1/3 times that in star connection, the starting torque will be reduced to one third. To get full torque in the motor it must be switched over to delta connection. A simple manual star-delta starter is shown in figure 8.4.

While making connections for star delta starting, care should be taken such that sequence of supply connections to the winding terminals does not change, while changing from star-connection to delta connection. Otherwise the motor will start rotating in the opposite direction when connections are changed from star to delta. Star-delta starters are available for manual operation using push button

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